# The Big Picture

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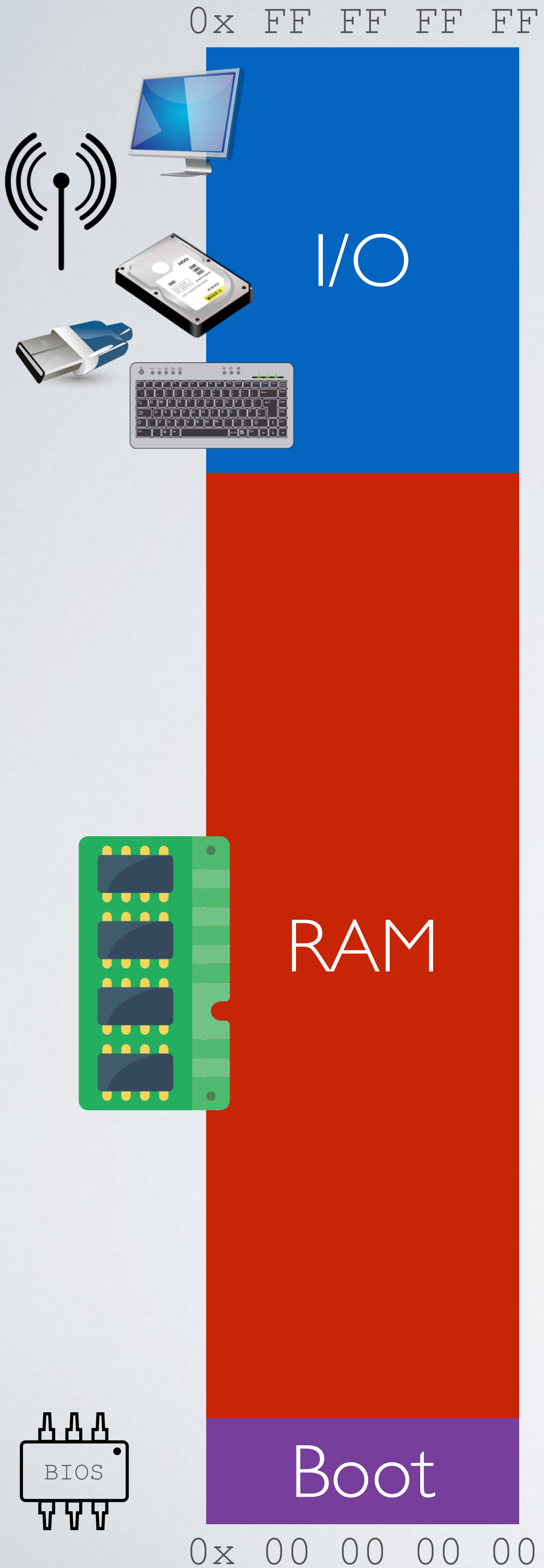
# Goals of this lecture

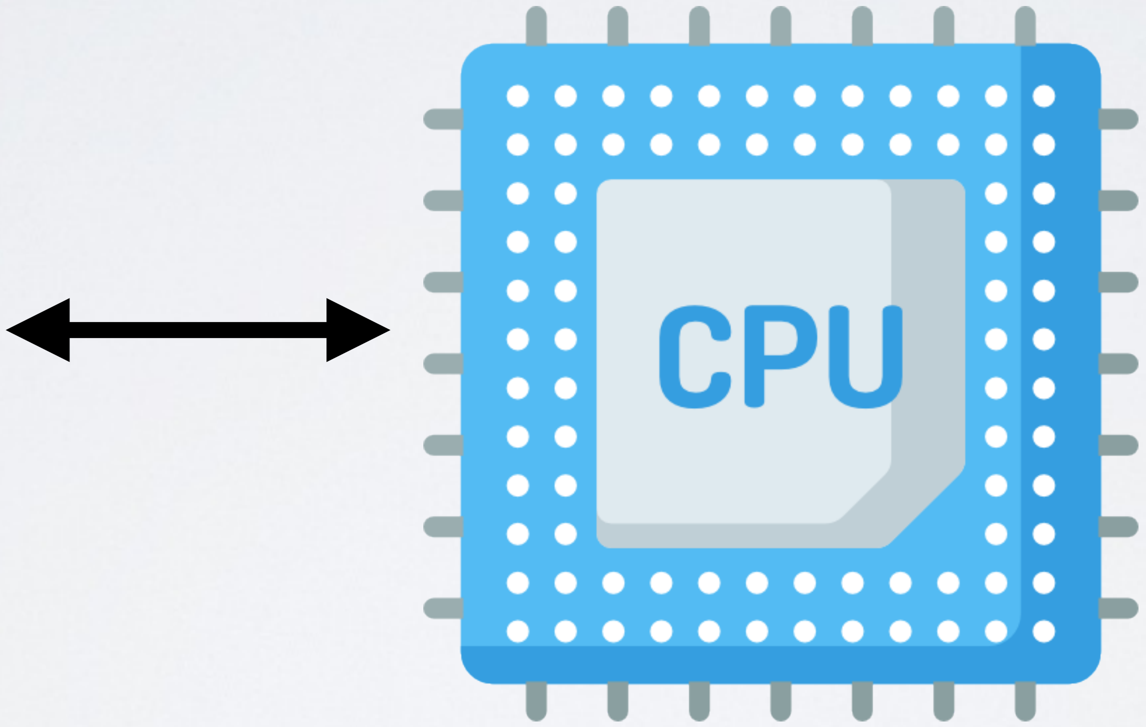
- Define what an Operating System is
- Explain how an OS works in a nutshell
- Bridge the gap between hardware (CSCB58) and systems programming (CSCB09)
- Give an overview of the course content and projects

# The big picture in 5 pieces

| | |
|---|---|
| The need for **bootstrapping** | |
| The need for **concurrency** | project 1 |
| The need for **user programs** | project 2 |
| The need for **virtual memory** | project 3 |
| The need for **a filesystem** | project 4 |

# Simple Computer Architecture
# Memory + CPU




for a more accurate and detailed map of the x86 memory look at https://wiki.osdev.org/Memory_Map_(x86)

# Each processor has its Instruction Set Architecture (ISA)

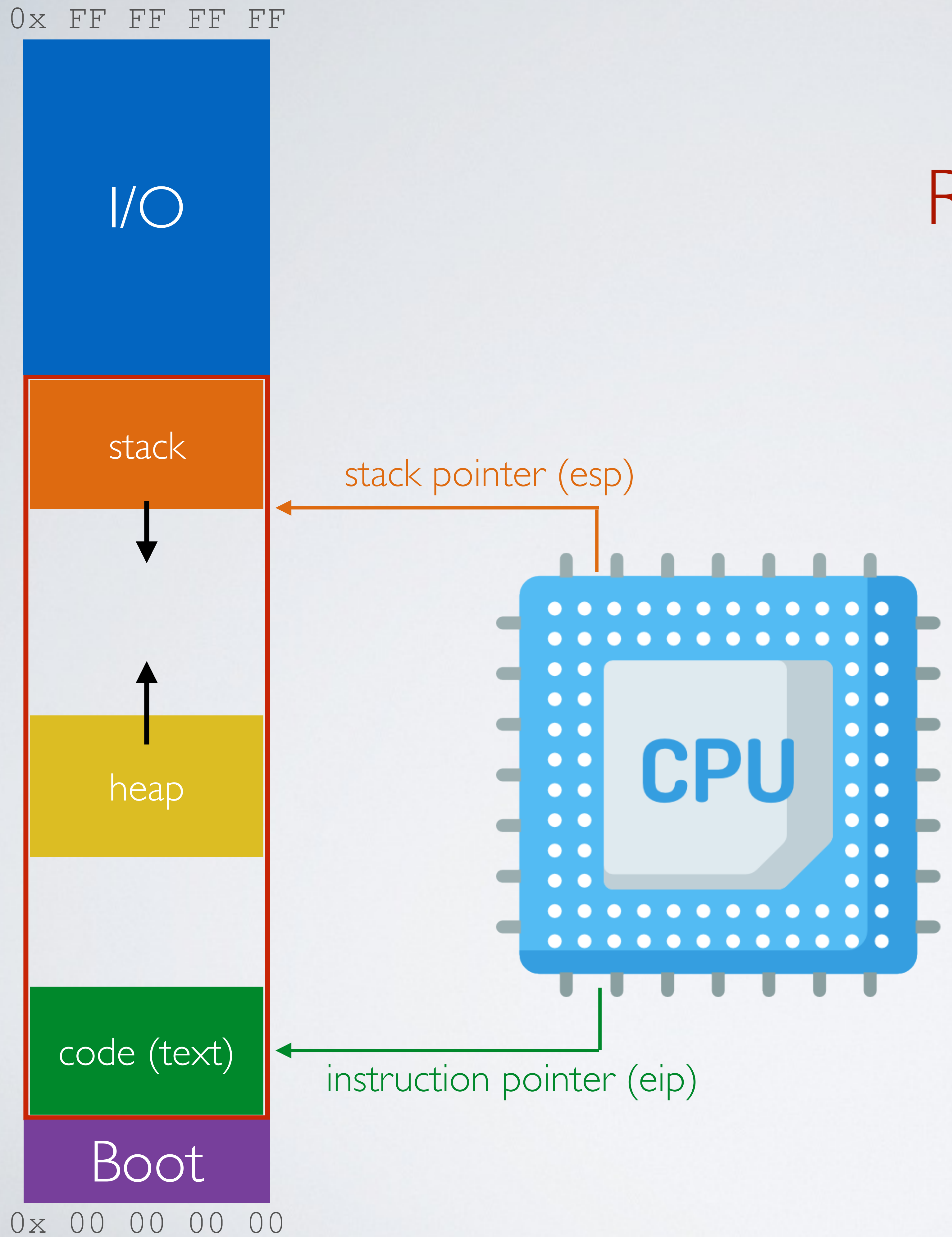
Processor executes instructions stored in memory

➡ Each instruction is a bit string that the processor understands as an operation

- arithmetic
- read/write bit strings
- bit logic
- jumps

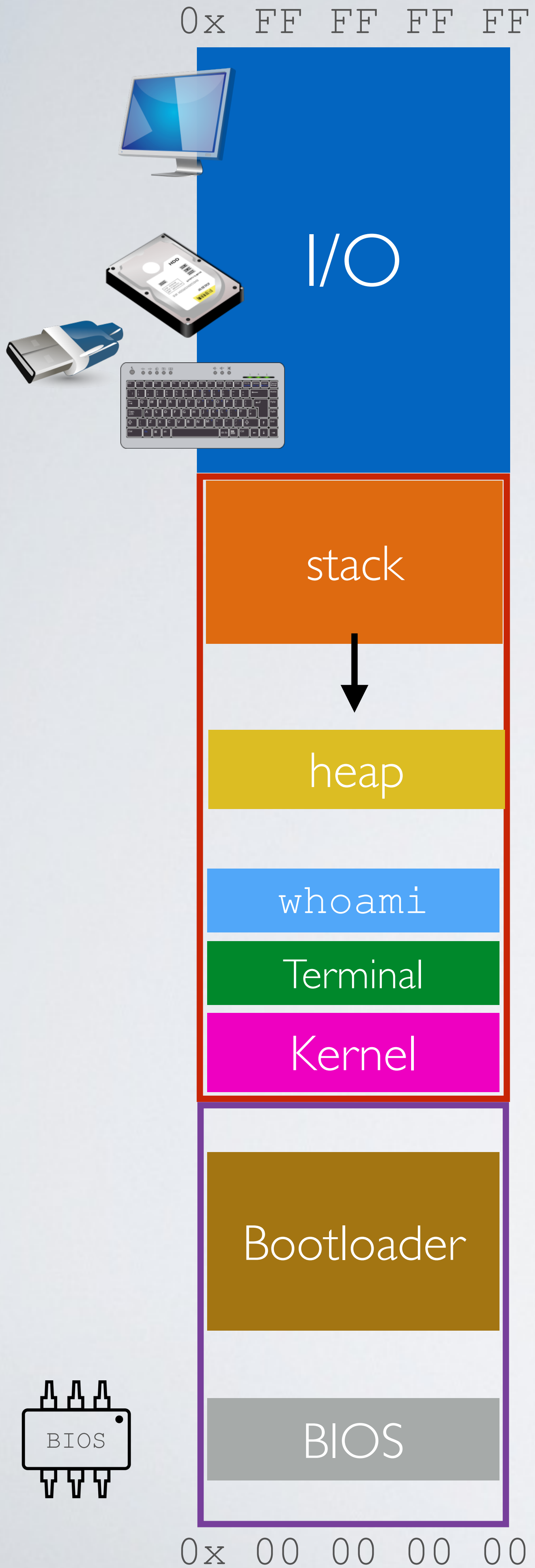
✓ ~2000 instructions on modern x86-64 processors

# Running one program



# The need for **bootstrapping**

# Bootstrapping




Step 5: using the terminal, users can execute programs (e.g Bash terminal) ... and repeat

Step 4: the kernel starts the user-interface program (e.g `Bash` terminal)

Step 3: the bootloader loads the OS kernel in RAM

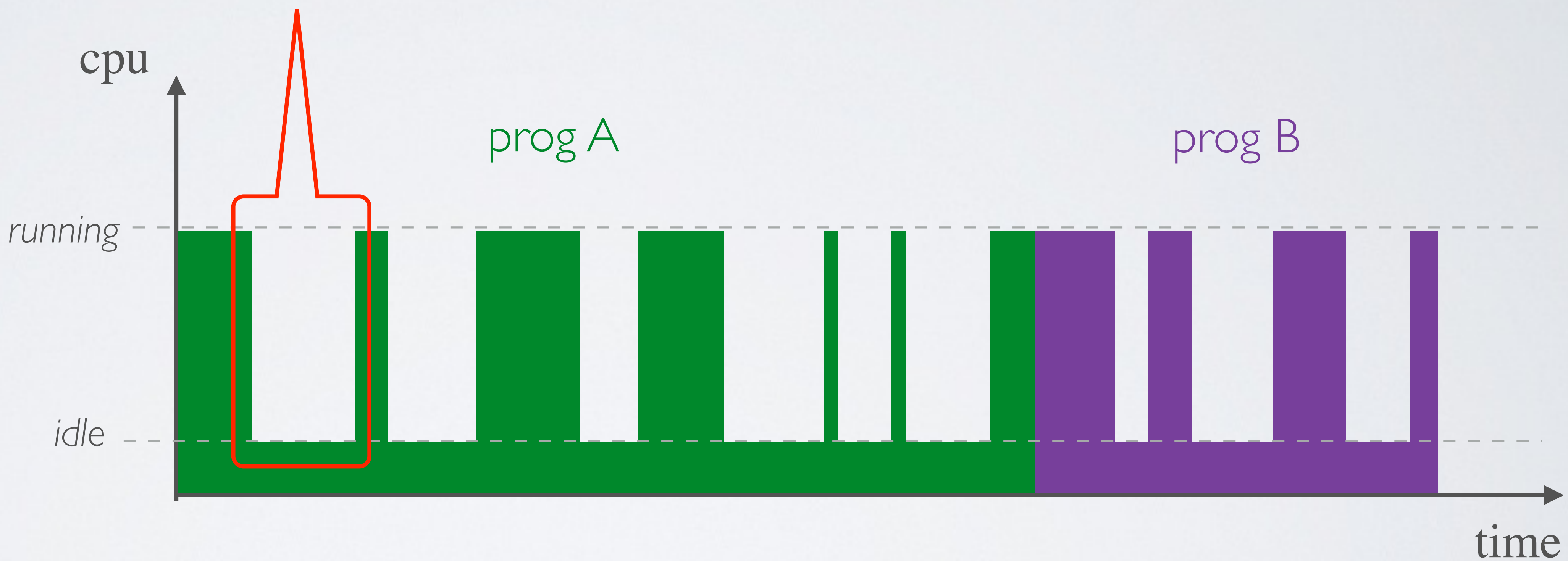
Step 2: the BIOS loads the **bootloader** from a device (hard-drive, USB, network ...) based on the configuration

Step 1: Power -on! The CPU starts executing code contained in the **BIOS** (basic input/output system)

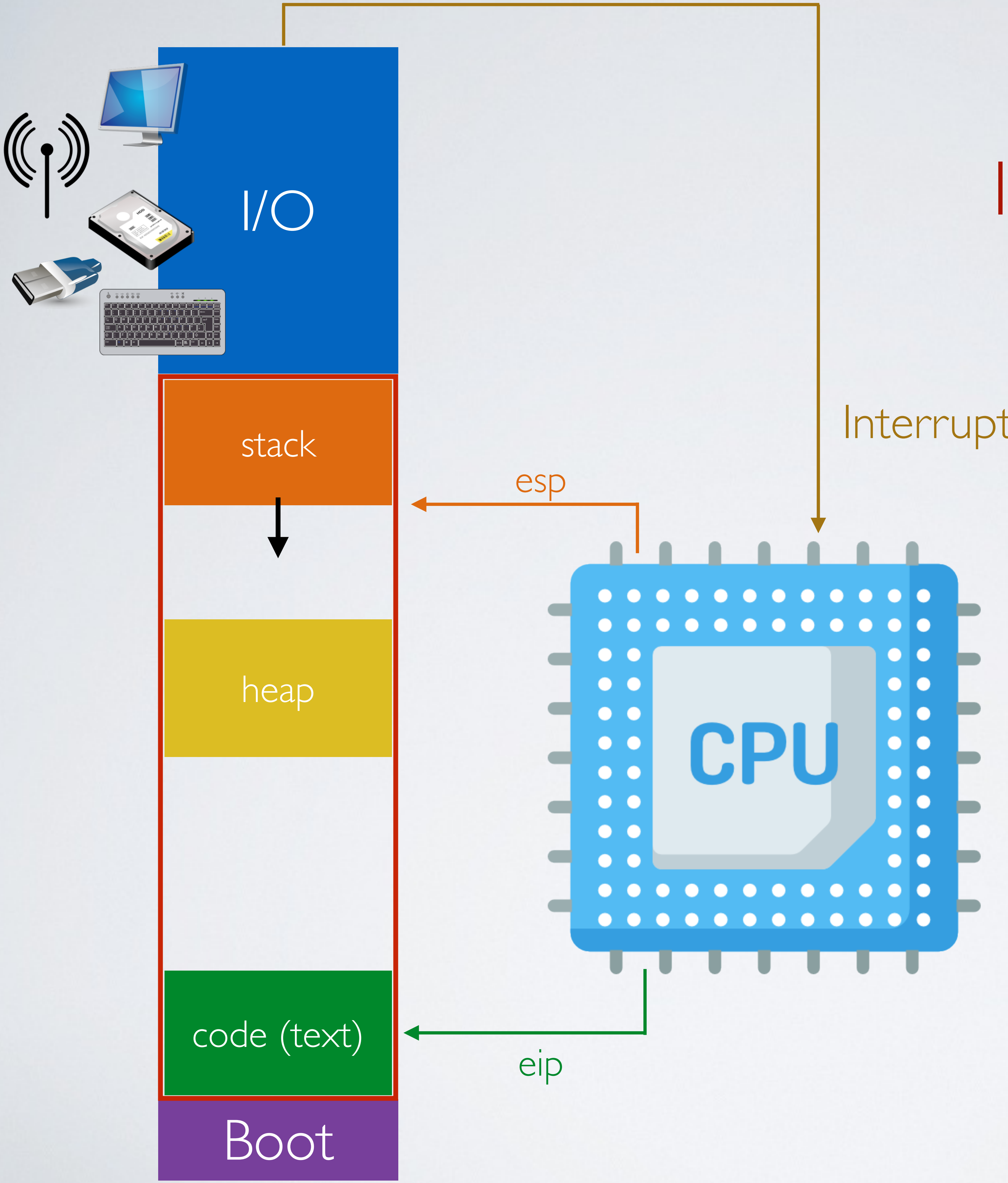
# The need for **concurrency**

# Running multiple programs one after the other


**Problem**: the CPU is waiting for I/O (polling)

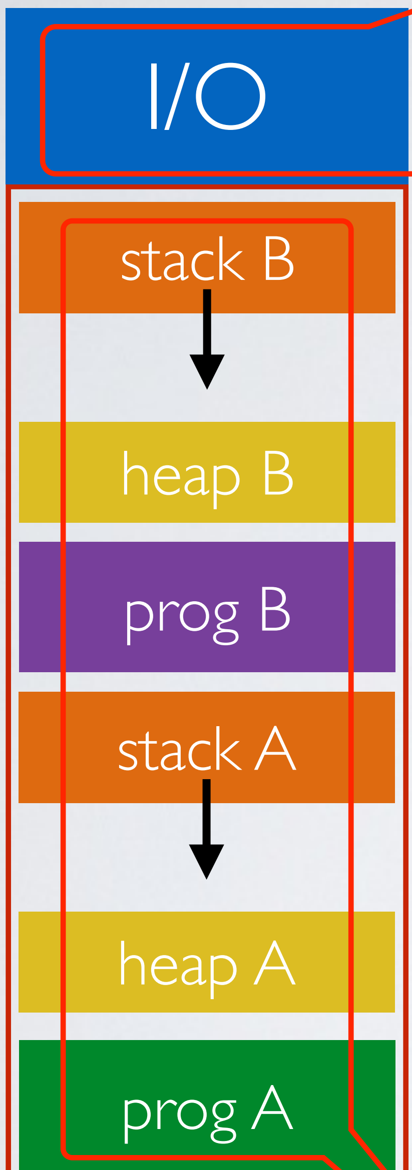


# I/O with interrupts

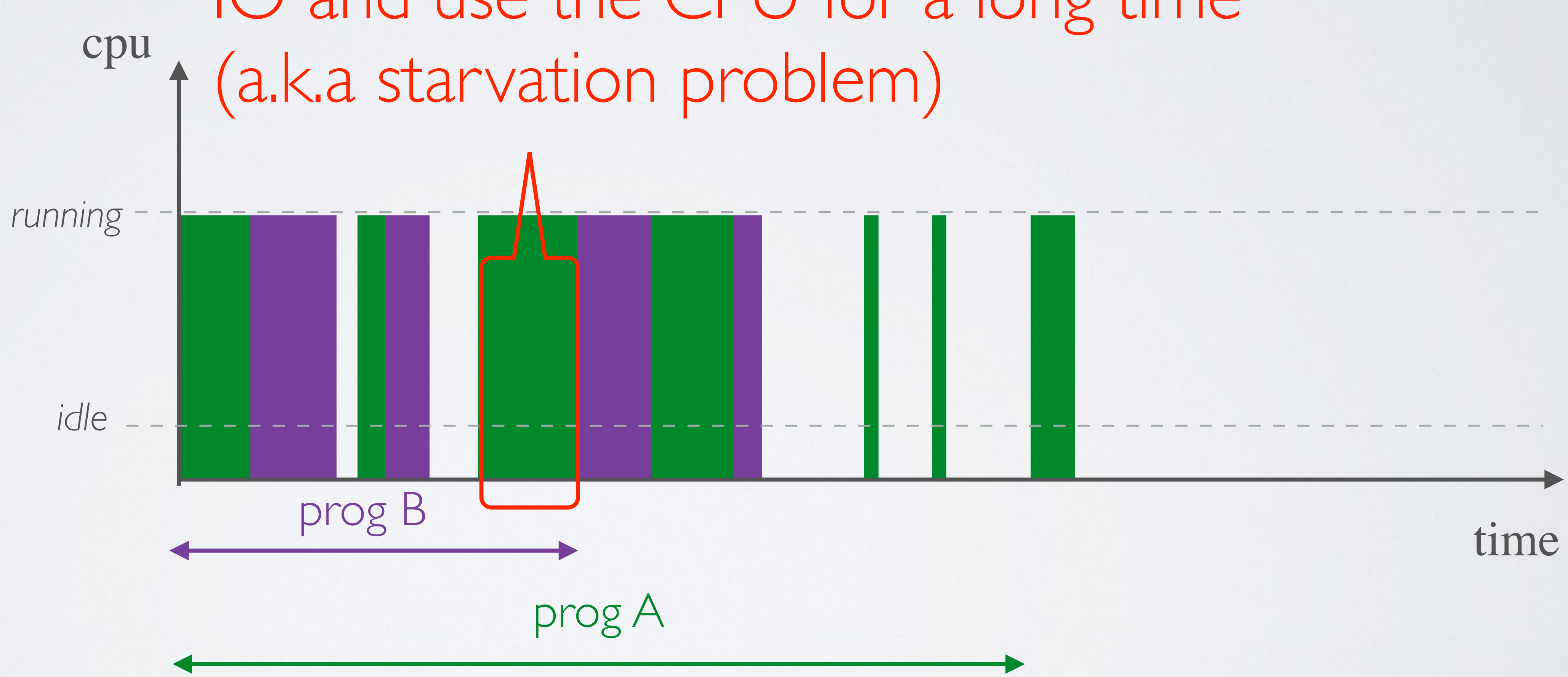


# Running multiple programs concurrently

**Problem**: concurrent access to I/O devices must be synchronized

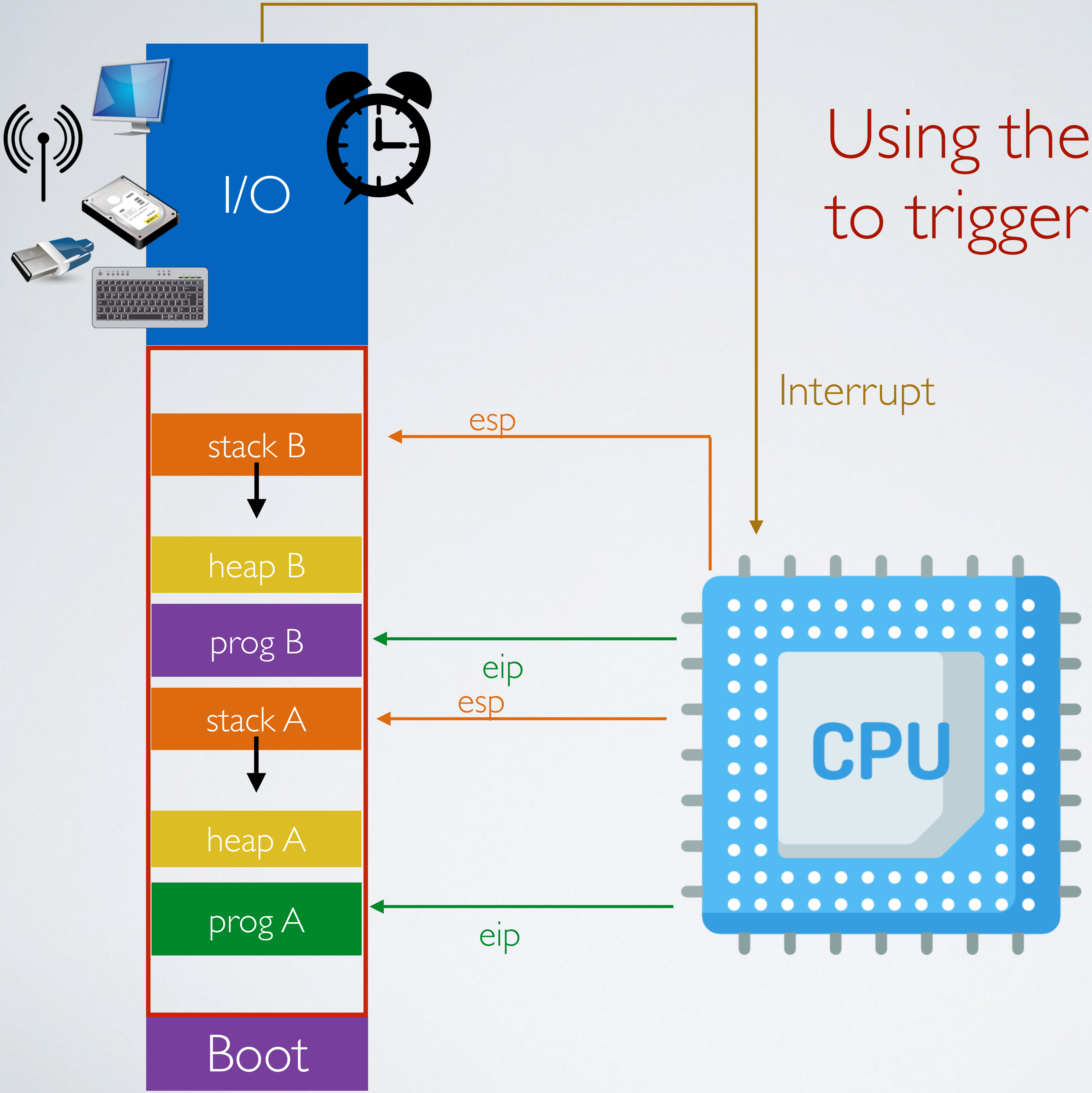



**Problem**: what if the program does not do any IO and use the CPU for a long time (a.k.a starvation problem)

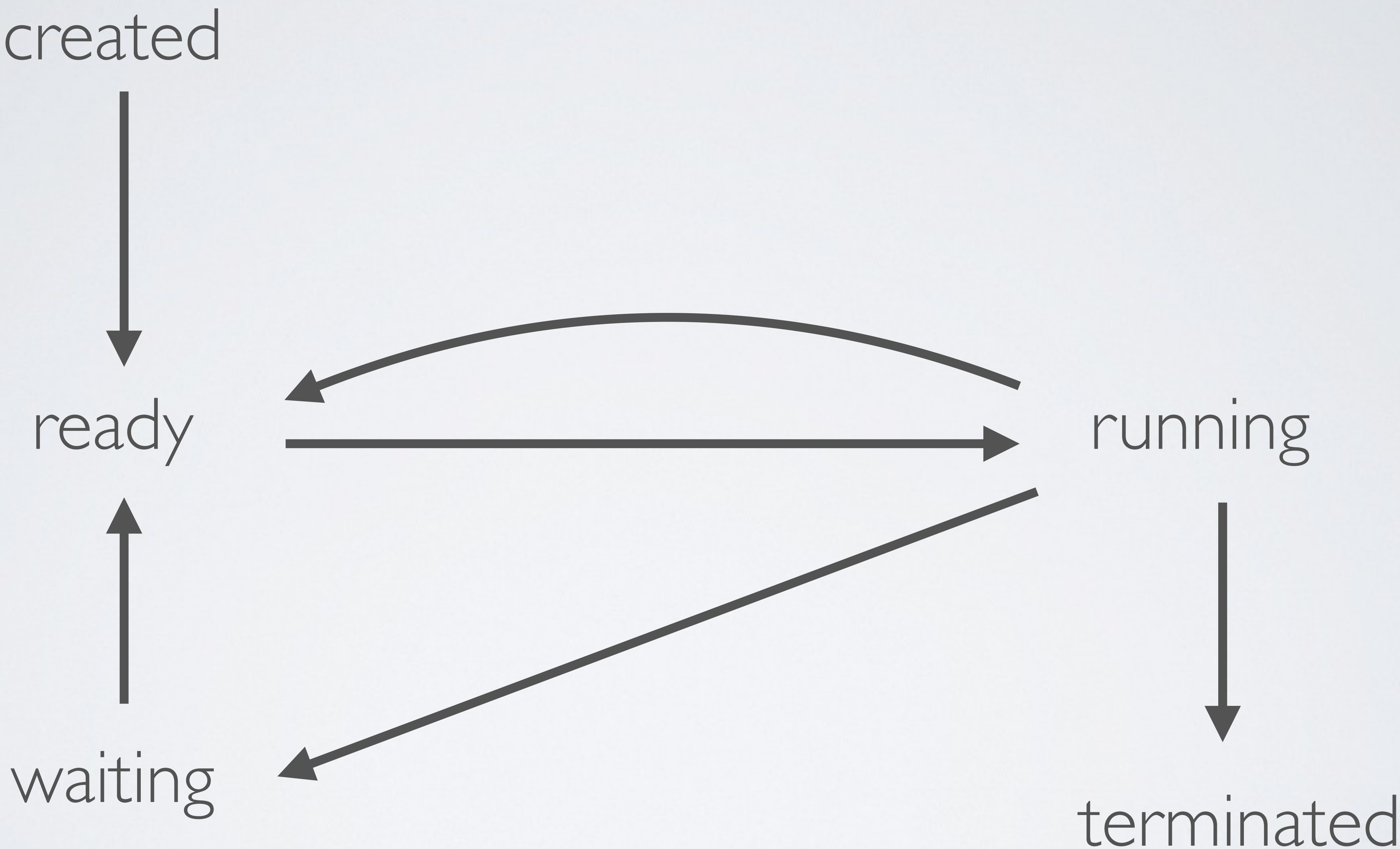



**Problem**: the programs and their stacks must co-exists in memory (coming next with virtual memory)

# Using the clock to trigger an interrupt



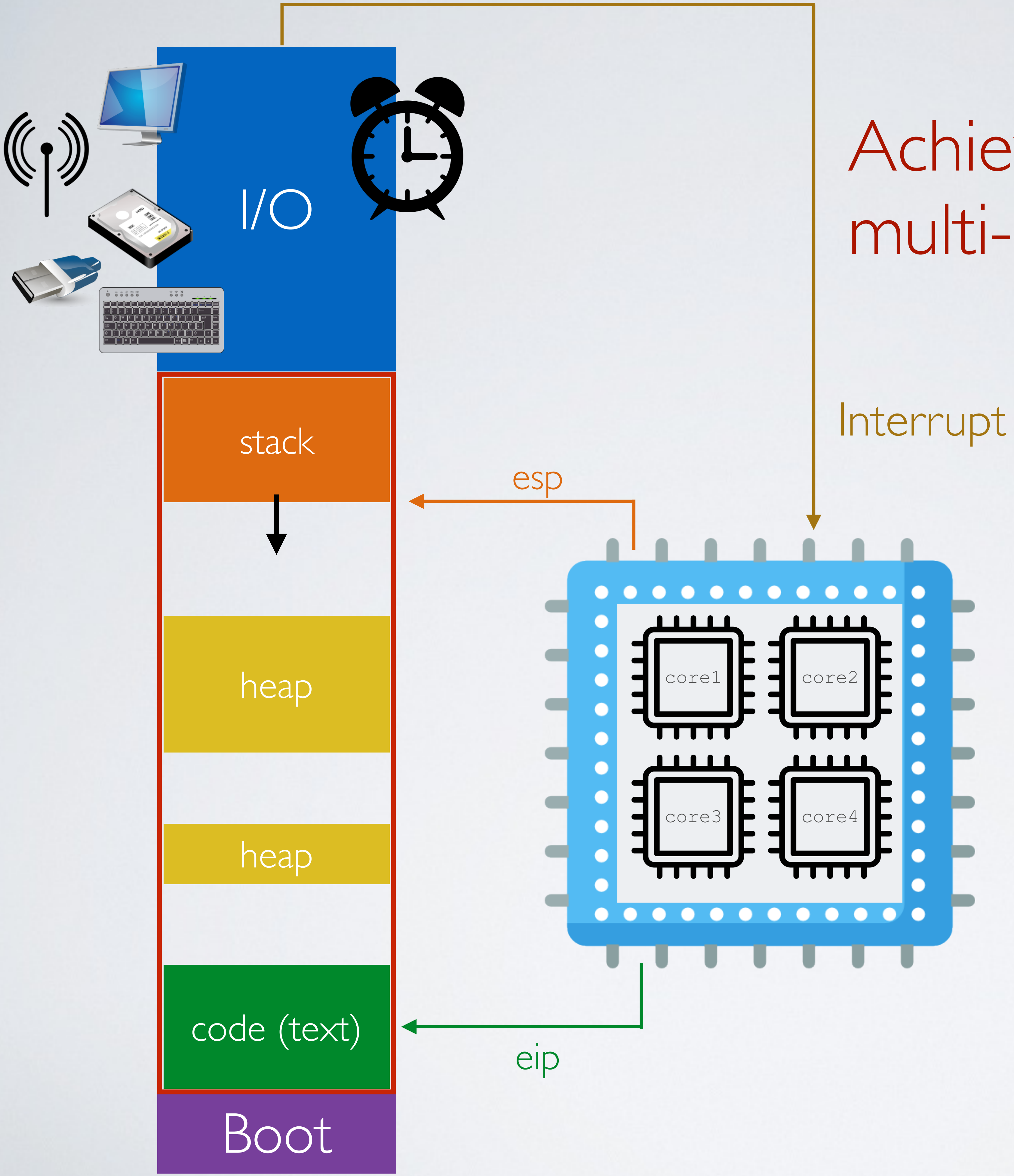
# Program States



# Other problems that we are going to address during the semester

- **Scheduling**
  Decide which process to execute when severals are ready to be run
- **Synchronization**
  Manage concurrent access to resources using semaphores, locks, monitors
- **Communication**
  Exchange messages between processes using IPC (sockets & signals)
- **User Threads**
  Lightweight concurrency within a process

# Achieving parallelism with multi-core processors



# The need for **user programs**

# The need for abstraction for user programs

How to write a user program like the *Bash* shell that reads keyboard inputs from the user?

➡ Read input data from the I/O device directly? But which one?

- The one connected to the PS2 port?
- The one connected to the USB?
- The one connected to the bluetooth?
- The remote one connected to the network?

◉ User programs do not operate I/O devices directly

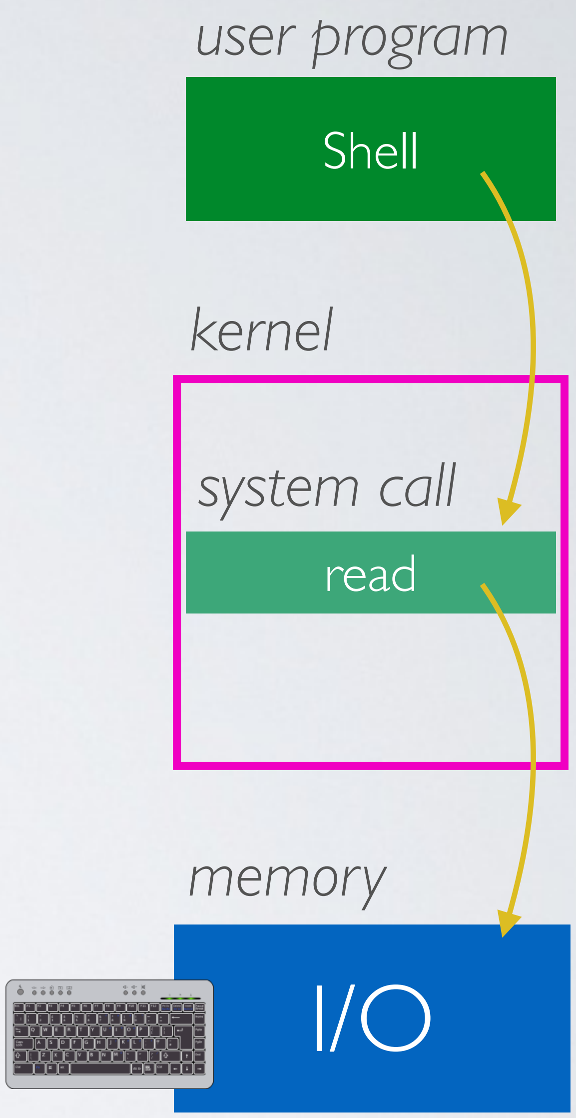
✓ The OS abstracts those functionalities and provide them as **system calls**

# System Calls

➡ Provide user programs with an API to use the services of operating system

There are 5 categories of system calls

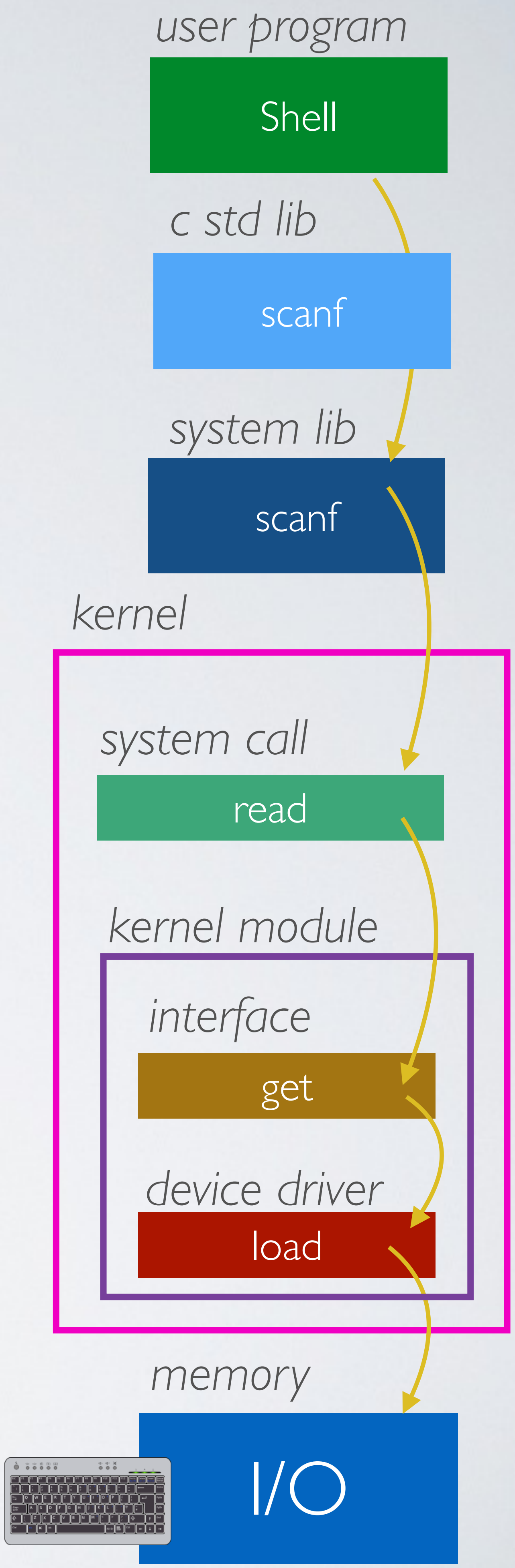
- Process control
- File management
- Device management
- Information/maintenance (system configuration)
- Communication (IPC)
- Protection




✓ There are about 500 system calls on Linux 7.0

# In reality, many (many) level of abstraction and modularity

➡ This is what makes developing OS very challenging (CSCB07)



# With concurrency

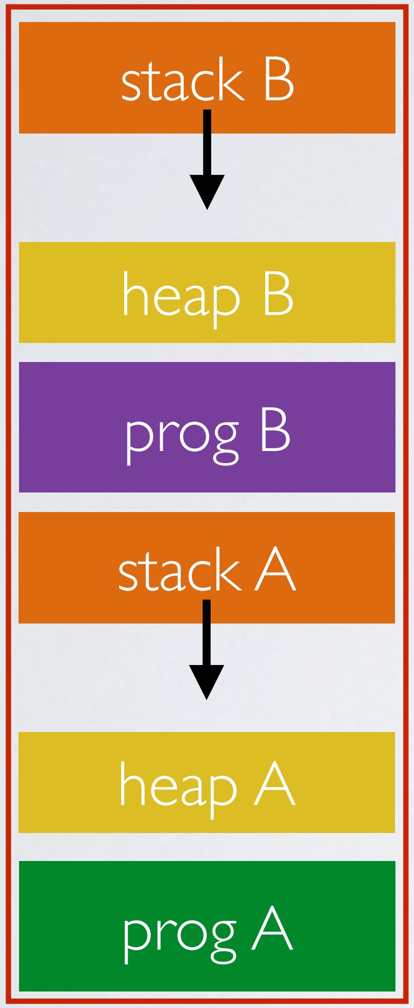
✓ **From the system perspective**
better CPU usage resulting in a faster execution overall (but not individually)

✓ **From the user perspective**
programs seem to be executed in parallel

➡ But it requires scheduling, synchronization and some protection mechanisms

# The need for **virtual memory**

# The problem of managing the memory




How to make programs and execution contexts co-exists in memory?

✓ Placing multiple execution contexts (stack and heap) at random locations in memory is not a problem ... ... well, as long as your have enough memory

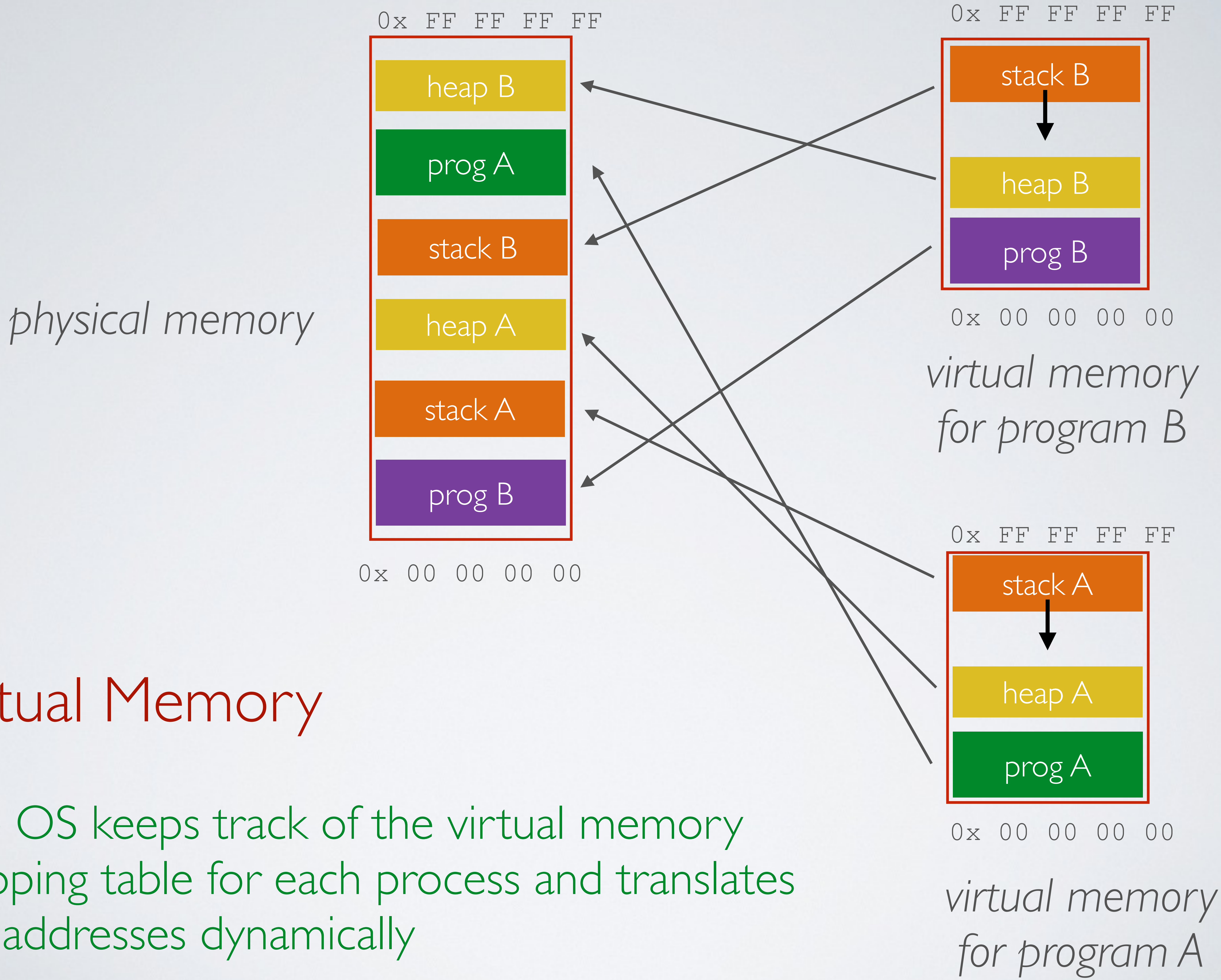
◉ However having programs placed at random locations is problematic

# Let's look at some C code and its binary

```c
#include <stdio.h>

int foo(){
    printf("hello world!");
}

int main(int argc, char **argv){
    foo();
}
```

```
0804840b <foo>:
 804840b:   55                      push   ebp
 804840c:   89 e5                   mov    ebp,esp
 804840e:   83 ec 08                sub    esp,0x8
 8048411:   83 ec 0c                sub    esp,0xc
 8048414:   68 d0 84 04 08          push   0x80484d0
 8048419:   e8 c2 fe ff ff          call   80482e0 <printf@plt>
 804841e:   83 c4 10                add    esp,0x10
 8048421:   90                      nop
 8048422:   c9                      leave
 8048423:   c3                      ret

08048424 <main>:
 8048424:   8d 4c 24 04             lea    ecx,[esp+0x4]
 8048428:   83 e4 f0                and    esp,0xfffffff0
 804842b:   ff 71 fc                push   DWORD PTR [ecx-0x4]
 804842e:   55                      push   ebp
 804842f:   89 e5                   mov    ebp,esp
 8048431:   51                      push   ecx
 8048432:   83 ec 04                sub    esp,0x4
 8048435:   e8 d1 ff ff ff          call   804840b <foo>
 804843a:   b8 00 00 00 00          mov    eax,0x0
 804843f:   83 c4 04                add    esp,0x4
 8048442:   59                      pop    ecx
 8048443:   5d                      pop    ebp
 8048444:   8d 61 fc                lea    esp,[ecx-0x4]
 8048447:   c3                      ret
 8048448:   66 90                   xchg   ax,ax
 804844a:   66 90                   xchg   ax,ax
 804844c:   66 90                   xchg   ax,ax
 804844e:   66 90                   xchg   ax,ax
```

Since function addresses and others are hard-encoded in the binary, the program cannot be placed at random locations in memory




# Virtual Memory

The OS keeps track of the virtual memory mapping table for each process and translates the addresses dynamically

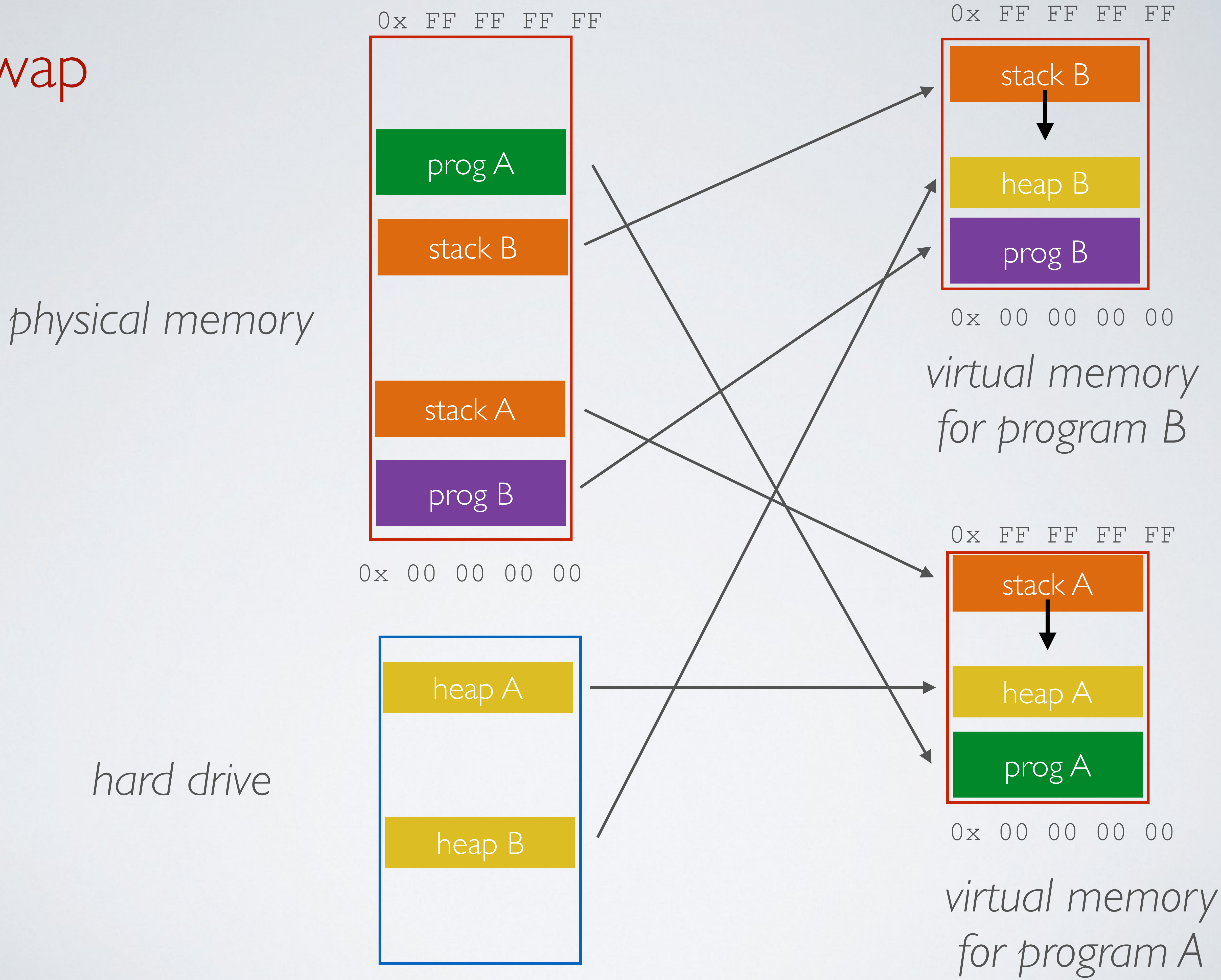
# Another problem

What if we run out of memory because of too many concurrent programs?

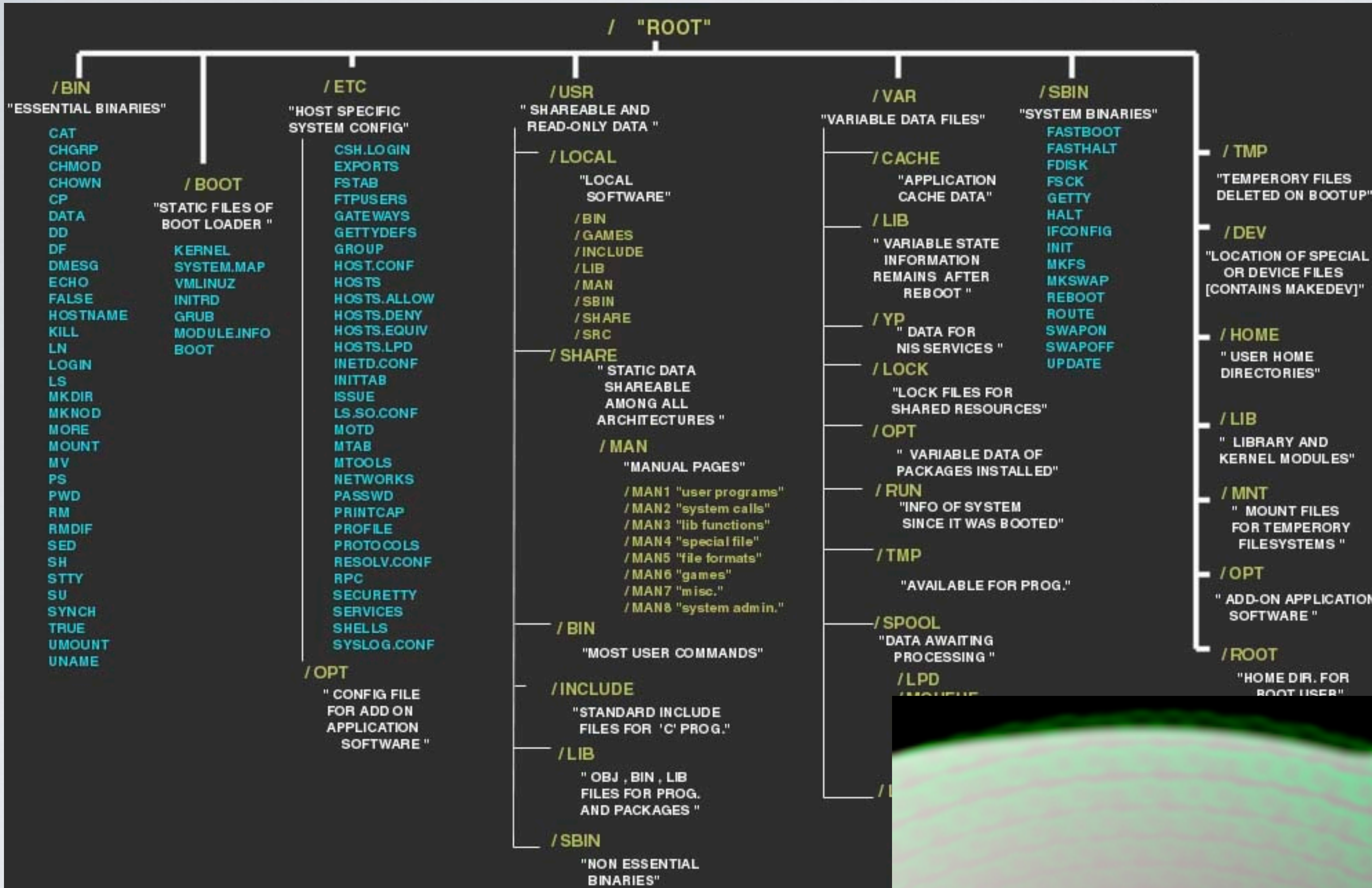
✓ Swap memory
move some data to the disk

➡ Managing memory becomes very complex
but necessary

# Swap


physical memory
0x FF FF FF FF
prog A
stack B
stack A
prog B
0x 00 00 00 00
hard drive
heap A
heap B
0x FF FF FF FF
stack B
heap B
prog B
0x 00 00 00 00
virtual memory for program B
0x FF FF FF FF
stack A
heap A
prog A
0x 00 00 00 00
virtual memory for program A

# The need for a **file system**




# Files and Directories

versus

Reality

So, what is an **operating system**?

# Operating System

➡ In a nutshell, an OS manages hardware and runs programs

- creates and manages processes
- manages access to the memory (including RAM and I/O)
- manages files and directories of the filesystem on disk(s)
- enforces protection mechanisms for reliability and security
- enables inter-process communication